Semantic Video Indexing by Fusing Explicit and Implicit Context Spaces*

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ABSTRACT

This paper addresses the problem of context-based concept fusion (CBCF) for concept detection and semantic video indexing. We introduce a novel framework based on constructing context spaces of concepts, such that the contextual correlations are used to improve the performance of concept detectors. Different from traditional CBCF approach, we present two kinds of such context spaces: explicit context space for modeling the correlation of pairwise concepts, and implicit context space for representing latent themes trained from a set of concepts. The final concept detection scores are then directly fused from explicit and implicit context spaces. Experiments are presented on TRECVid 2006 benchmark and the comparisons with several state-of-the-art approaches demonstrate the effectiveness of proposed framework.

Categories and Subject Descriptors: I.2.10 [Artificial Intelligence]: Vision and Scene Understanding—*video analysis*; H.3.1 [Information Storage and Retrieval]: Content Analysis and Indexing—*indexing method*.

General Terms: Algorithms, Experimentation, Performance.

Keywords: Semantic video indexing, context-based concept fusion, context space, sparse coding, TRECVid.

1. INTRODUCTION

As huge amount of web video collections, broadcast news and documentary video archives are generated and shared in the Internet, many works have been done for efficiency and effectiveness video indexing and retrieval in recent years. In this paper, we study the problem of concept detection and semantic video indexing. That is, to detect a number of semantic concepts in video clips and provide concept-level access. These semantic concepts include object (e.g. car), scene (e.g. cityscape), and event (e.g. people dancing). Recent works on the high-level feature extraction [10] of TRECVid [9] demonstrate great progress for this problem. Two aspects are mainly utilized for the improvement. On one hand, concept detection accuracy is improved based on the evolution of image classification techniques; more precision approach for modeling the video clips and keyframes are employed, e.g. local features [4] and color descriptors [13]. On the other hand, large-scale concept sets such as LSCOM [7] are built for multimedia retrieval research.

Based on the video content modeling and large-scale concept set, some concept detection frameworks including VIREO-374 [4], Columbia-374 [16] and Mediamill-101 [12] are proposed and achieve start-of-the-art results. One assumption is that detection scores from different concepts are independent to each other and our foucs is to explore semantic knowledge among concepts for video indexing. Particulary, we aim to develop a context-based concept fusion (CBCF) framework to enhance the concept detection results.

In this paper, we introduce a novel framework based on constructing context spaces of concepts, such that the contextual correlations are used to improve the performance of concept detectors. The correlations are modeled by the fusion of two perspectives. The first perspective is to utilize pairwise concepts' relation directly, which we denote as "Explicit Context Space". For example, the concept "airplane" frequently co-occurs with concept "sky". The second perspective is to learn a new latent theme set (i.e. "Implicit Context Space") and map concept scores to the context space. For example, the concept "airplane" may relate to the theme representing "airplane in the sky" or "airplane in the airport" closely. Specifically, the implicit context space is based on the sparse codes from original concept detection scores for it is able to learn over-complete basis sets, which means the number of basis can be larger than the number of concept detectors. To the best of our knowledge, there is no existing work that fuse this two perspectives for context-based concept fusion. Experiments on TRECVid 2006 benchmark [9] and VIREO-374 baseline [4] demonstrate the effectiveness of proposed framework.

The remainder of this paper is organized as follows. Section 2 reviews the related literature on context-based concept fusion. Section 3 describes our framework based on explicit and implicit context spaces. In Section 4 we demonstrate the experiment setup and results on TRECVid dataset. And finally, we conclude our work in Section 5.

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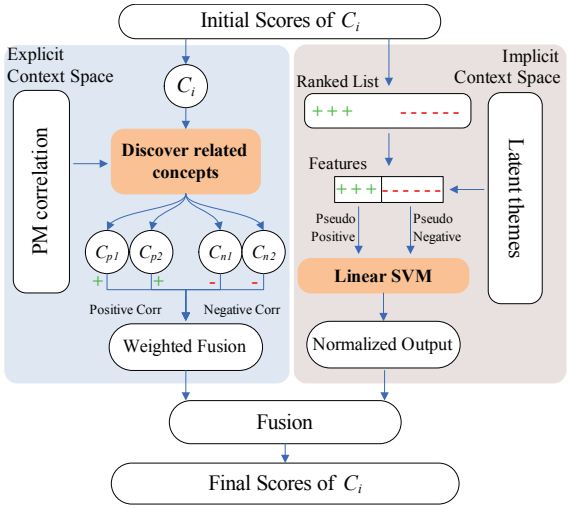

Figure 1: Our CBCF framework for concept C_i.

2. RELATED WORKS

Many different context-based concept fusion approaches exist in the literature [1, 2, 3, 5, 8, 11, 14, 15, 17] and we summarize them into two categories. The first one is based on exploration of pairwise concept correlation [3, 14, 17], which is closely related to our explicit context space. In [17], concurrent relation is explicitly modeled by a concurrent matrix to refine the annotations. Jiang et al. [3] propose domain adaptive semantic diffusion (DASD) by considering both semantic context and domain adaptation ability. Probably most similar to ours is the work of Wei et al. [14] on exploring inter-concept relationship with contextual space. Different with their approach, our explicit context space is simpler and more efficient.

In addition to the pairwise concept correlation, other approaches for CBCF are two-layer learning techniques [1, 5, 11]. Smith et al. [11] proposed the Discriminative Model Fusion (DMF) method for multimedia semantic indexing; the outputs of all the individual detectors are aggregated as the features of SVM. Jiang et al. [1] proposed an active CBCF method based on the annotation of a small number of concepts by users. Then context-based SVM is learnt to improve the detection score. In [5], detection scores of 75 concepts with highest mutual information with the target are chosen to form the feature vector for each concepts. Our implicit context space also belongs to the two-layer learning techniques but can achieve better detection results.

3. APPROACH

Given a semantic lexicon of m concepts $C = \{c_1, c_2, ..., c_m\}$ and a video dataset of n shots $X = \{x_1, x_2, ..., x_n\}$, concept detection aims to give prediction scores of the concepts to each shot. We define the ground-truth label and prediction score of video shot k for concept i as y_{ik} and s_{ik}, respectively. And the goal of context-based concept fusion is to generate the refined score s'_{ik} that improves concept detection results.

As illustrated in Figure 1, our CBCF framework contains two phases. Explicit and implicit context spaces are independently applied and then fused to get final results.

3.1 Explicit Context Space

We start by describing the explicit context space. Explicit context space is used to represent semantic relation of pairwise concepts. Here we construct the correlation graph based on the annotation of training data. Similar to previous works [3, 14], Pearson product moment correlation (PM) is applied to model the correlation:

$$PM(c_i, c_j) = \frac{\sum_{k=1}^{n}(y_{ik} - \mu_i)(y_{jk} - \mu_j)}{(n-1)\sigma_i\sigma_j}$$

where $y_{ik} = 1$ indicates shot k annotated with concept c_i, otherwise $y_{ik} = 0$; μ_i and σ_i are the mean and standard deviation of $\{y_{ik}|k = 1, 2, ..., n\}$, respectively. The PM value ranges between +1 and -1.

Different from the post-processing on PM matrix (e.g., semantic diffusion in [3] and spectral decomposition in [14]), we directly enhance detection scores by weighted fusing scores of the most P positive- and N negative-correlated concepts:

$$s^E_{ik} = \sum_{j \in C_p} w_{ij} s_{jk} + \sum_{j \in C_n} w_{ij} s_{jk}$$

where C_p and C_n denote the set of positive- and negative-correlated concepts, and $w_{ij} = \exp(PM(i, j))$. The explicit context space is simple and efficient; yet it can achieve comparable results to other approaches.

3.2 Implicit Context Space

In this subsection, we explain the implicit context space for context-based concept fusion. Besides the selected concept subset for representing pairwise correlation, knowledge from a large concept lexicon can be useful for enhancing individual concept detection. It is intuitive to form the feature vector by applying detection scores of whole lexicon or a part of them for each shot, i.e. the two-layer learning techniques [5, 11]. However, noise or unreliable first-layer detection results may influence prediction of the second-layer.

Let $s_k = [s_{1k}, s_{2k}, ..., s_{mk}]^T$ be the detection scores of shot k in the concept lexicon C. Our method aims to find some "latent theme" with the ability to model relation of concepts. We begin from vector quantization of feature vectors $S = \{s_k|k = 1, 2, ..., n\}$. Intuitively k-means clustering may be applied to resolve the problem. However, the hard-assignment would assign s_k to only one of the clusters and may not capture enough context information. Instead, we use the sparse coding algorithm for constructing implicit context space. It is worth noting there are three advantages that may benefit. Sparse coding can return the basis vectors that capture higher-level information in the data. Furthermore, over-complete basis sets are learned and the number of basis may larger than the number of concept detectors. Finally, the feature vector s_k is assigned to several basis and simultaneously the transformed feature vectors are sparse. Learning sparse codes can format as the following optimization problem:

$$\begin{aligned} \min_{B,f} \quad & \sum_{k=1}^{n} ||s_k - Bf_k||_2^2 + \lambda ||f_k||_1 \\ s.t. \quad & \sum_i B_{ij}^2 \leq 1, \ \forall j = 1, 2, ..., n \end{aligned}$$

where B is $m \times d$ matrix, column vectors of B are the sparse coding basis, d is the number of basis, f_k is the transformed feature vector for shot k, and $||\cdot||_q$ denotes the L_q norm

Figure 2: TRECVid 2006 high-level features.

$(q = 1, 2)$. In our implementation, a feature subset S_0 is randomly selected from S for learning the basis and we solve the optimization problem using an efficient algorithm [6].

SVM with linear kernel is selected as the second-layer learner. In the training stage, a ranked list of scores is firstly obtained for a given concept c_i. We select the top n_p shots to form pseudo-positive set and randomly select n_n shots to form the bottom half of list as the pseudo-negative examples. The transformed features f_k corresponding to shot x_k are the input of SVM. In the testing stage, each shot x_k is given a prediction p_{ik}. The final results are the normalized output of predictions:

$$s_{ik}^{I} = \frac{p_{ik} - \mu_i^p}{\sigma_i^p}$$

where μ_i^p and σ_i^p are the mean and standard deviation of $\{p_{ik}|k = 1, 2, ..., n\}$, respectively.

3.3 Fusion

Explicit and implicit context spaces are with complementary ability for context-based concept fusion. The explicit context space utilizes information from training set, i.e. the annotation for PM correlation, while implicit context space utilizes from testing set, i.e. first-layer detection scores. In addition, the explicit space is directly fusing several related concepts, while the implicit space generates latent themes by universal set of concept lexicon. To obtain an better detection which combines the two context spaces, we apply the average fusion strategy:

$$s'_{ik} = s_{ik} + \alpha \cdot s_{ik}^{E} + \beta \cdot s_{ik}^{I}$$

Experiments in the next section will validate the effectiveness of the approach.

4. EXPERIMENTS

4.1 Experiment Setup

We examine our CBCF framework on TRECVid dataset [9] and take publicly available VIREO-374 detectors [4] as our baseline. The concept detectors are trained using TRECVid 2005 development set with 43,873 shots and evaluated in TRECVid 2006 testing set with 79,484 shots. There are 374 concept detectors in VIREO-374, and 20 of them are selected for official evaluation in TRECVid 2006 (Figure 2).

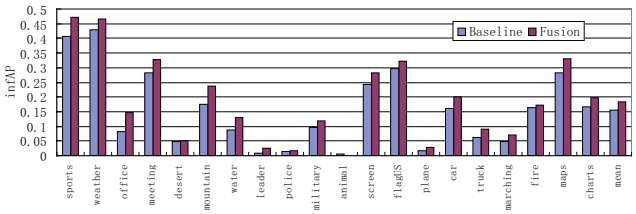

Figure 3: Performance comparison for each concept using VIREO-374 baseline and proposed method.

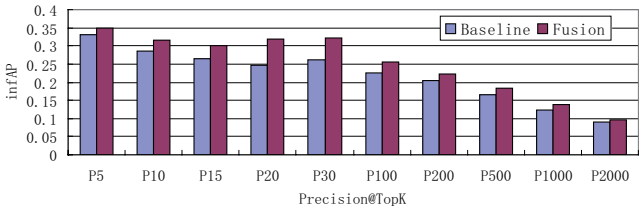

Figure 4: Distribution of true-positive shots (precision@k).

The detectors are learned using SVM and three individual visual features, i.e. grid-based color moments, wavelet texture, and bag-of-visual-words. The performances are measured by mean inferred average precision (infAP). Let R be the number of true-relevant shots in a set of size S and R_k be the number of relevant shots in the top k ranking shots. Let $I_k = 1$ if the kth shot is relevant and 0 otherwise. The average precision is defined as AP$=\frac{1}{R}\sum_{k=1}^{S}\frac{R_k}{k}\cdot I_k$. We adopt $S = 2000$ in our experiments.

4.2 Experiment Results

We compare our framework with the VIREO-374 [4] baseline, whose infAP achieves 0.1540. The comparison is illustrated in Figure 3. Among 20 evaluated concepts, the performances of 19 concepts are improved compared to VIREO-374 baseline and the infAP of our method is 0.1842. To confirm our observations, we show the overall distribution of true-relevant shots in Figure 4 and we find that our method improves the precision at different k. We will detail each components in the following parts.

4.2.1 *Effect of Explicit Context Space*

We empirically set correlated concept set size $P = |C_p| = 5$ and $N = |C_n| = 2$ for explicit context space. The infAP for similar approaches such as semantic diffusion [3] and context space [14] are 0.178 and 0.179, respectively. Result of our explicit context space is 0.1780, which is comparable with them.

4.2.2 *Effect of Implicit Context Space*

We adopt a two-layer SVM approach [5] with infAP 0.167[1] for comparison. Additionally, we implement the method that uses scores from 374 concepts and linear SVM but does not learn the latent themes (infAP 0.169). In order to show the effectiveness of sparse coding for implicit context space, we also apply the principle component analysis (PCA) to construct the space.

[1]This result is reported in [14].

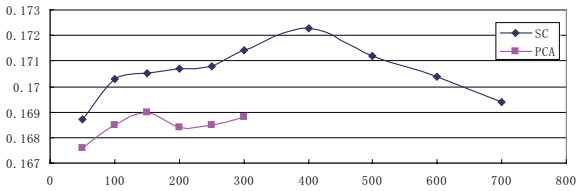

Figure 5: Sparse coding and PCA with different basis number for implicit context space.

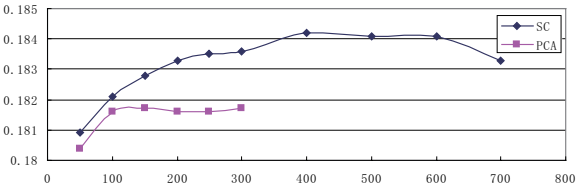

Figure 6: Effect of context spaces fusion.

The parameters used for sparse coding in Section 3.2 are set as: $\lambda = 1$, $|S_0| = 5000$ (also for PCA), $n_p = 60$, and $n_n = 240$. Comparisons with PCA are shown in Figure 5. We see that the curve of sparse coding is always over that for PCA and two-layer traditional SVM approaches. The highest infAP is reached when the number of latent themes is around 400, which indicates the advantage of over-complete basis in sparse coding. Another observation is that implicit context space tends to return more relevant shots compared with explicit context space (Table 1).

We also compare sparse coding and PCA after average fusion with explicit context space. The parameters α and β in Section 3.3 are simply set to 0.5. From Figure 6, we can clearly see the superiority of fusion with sparse coding over that with PCA. Overall, we can also observe the complementary ability of explicit and implicit context space and improvement of their fusion in both returned relevant video shots and infAP.

The only computationally expensive step of our framework is to learn the sparse codes in implicit context space construction, which takes about 20 minutes on a Intel Core 2 Duo 2.34GHz PC with 2G memory.

4.2.3 Comparison with State-of-the-Art

We compare our approach to three state-of-the-art approaches with VIREO-374 baseline for the semantic video indexing task, i.e., contextual correlation [15], context space [14], and domain adaptive semantic diffusion [3]. As shown in Table 2, the improvement of our explicit and implicit context space framework is 19.6%, which outperforms these approaches.

5. CONCLUSION

In this paper, we have introduced fusion of context spaces, a CBCF framework for Semantic Video Indexing. Different from previous CBCF approach, our context spaces include two elements: explicit context space for the correlation of pairwise concepts, and implicit context space for the latent themes learned using sparse coding. The framework is intuitive, effective, and easy to implement. Experiment results on TRECVid dataset and VIREO-374 baseline indicate the complementary ability of the two context spaces, and show the superiority over state-of-the-art approaches.

Table 1: Number of returned relevant shots for 20 concepts.

Total	Baseline	ECS	ICS	Fusion
9074	3561	3709	3777	3801

Table 2: Performance comparison with other works.

Methods	[14]	[15]	[3]	Ours
Improvement	15.3%	16.7%	17.5%	**19.6%**

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